Lecture Notes: General Relativity

Part I: Chapter 1 - Fundamentals of Topology

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Contents

Chapter 1

Fundamentals of Topology

When modeling physical events in spacetime or analyzing continuous systems, relying purely on distance (metrics) can be overly restrictive. Topology provides a deeper, more general framework by focusing purely on the concept of "open sets." This powerful abstraction allows us to define continuity and spatial deformation without needing a rigid concept of distance.

1.1 Axiomatic Definition of a Topological Space

Let X be a set (for instance, a continuous collection of physical events in spacetime). A topology on X is a collection $T = \{U_\alpha \subseteq X \mid \alpha \in \Lambda\}$ of subsets of X, called **open sets**, that strictly satisfies three fundamental axioms:

1. **Trivial Sets:** The empty set and the entire space are inherently open.

$$\emptyset, X \in T$$

2. **Finite Intersections:** The intersection of any finite number of open sets is open.

$$U_1, U_2 \in T \implies U_1 \cap U_2 \in T$$

3. **Arbitrary Unions:** The union of any collection of open sets is open.

$$U_\alpha \in T \implies \bigcup_\alpha U_\alpha \in T$$

1.2 Problem: Expressing and Expanding to Derive a Topology

To avoid the cumbersome process of explicitly listing every single open set in a massive space, we can mathematically "express and expand" a topology using smaller, fundamental building blocks. This is executed in two logical steps using a **Prebasis** and a **Basis**.

- **Step 1 (Expressing the Basis from a Prebasis):** We start with a raw collection of subsets (a prebasis). By taking all possible *finite intersections* of these subsets, we derive a **Basis** $B = \{V_\alpha \subseteq X\}$. The basis must satisfy two structural conditions:
  1. It completely covers the space: $\bigcup V_\alpha = X$.
  2. For any two basis elements $V_1, V_2 \in B$ and a point $x \in V_1 \cap V_2$, there exists a third basis element $V_3 \in B$ such that $x \in V_3 \subseteq V_1 \cap V_2$.
- **Step 2 (Expanding the Basis to the Topology):** We then take all *arbitrary unions* of these basis elements. Thus, a set U is defined as open ($U \in T$) if and only if it can be written as a union of basis elements:

$$U \in T \iff U = \bigcup V_\alpha$$

Logical Meaning: This two-step process allows us to systematically eliminate redundant subsets. By defining just the fundamental "bubbles" (the basis), we automatically generate the entire, rigorously complete topological structure.

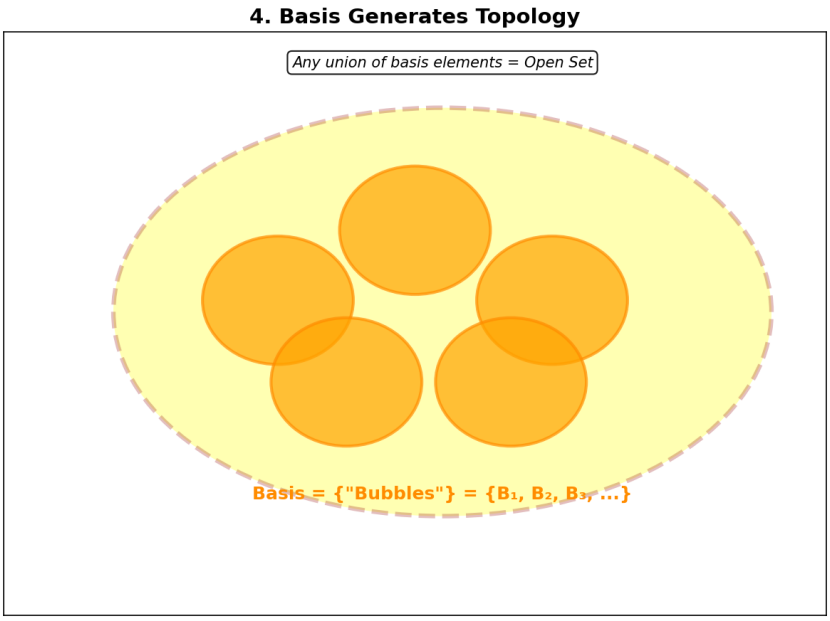


Figure 1.1: Basis generates Topology: Arbitrary union of basis elements = Open Set

1.3 The Usual Topology on $\mathbb{R}$ and $\mathbb{R}^n$

We can directly apply the basis concept to generate the "usual topology" for standard real number spaces.

For 1D Space ($\mathbb{R}$): The basis B consists of all open intervals (a, b). A set U is open if every point has a small interval around it that stays inside U:

$$U \in T \iff \forall x \in U, \exists (a, b) \subseteq U \text{ s.t. } x \in (a, b)$$

paragraphFor N-Dimensional Space ($\mathbb{R}^n$): We generalize the open interval to an N-dimensional open ball, $B(x, r)$, centered at x with radius r.

$$U \subseteq \mathbb{R}^n \text{ is open} \iff \forall x \in U, \exists r > 0 \text{ s.t. } B(x, r) \subseteq U$$

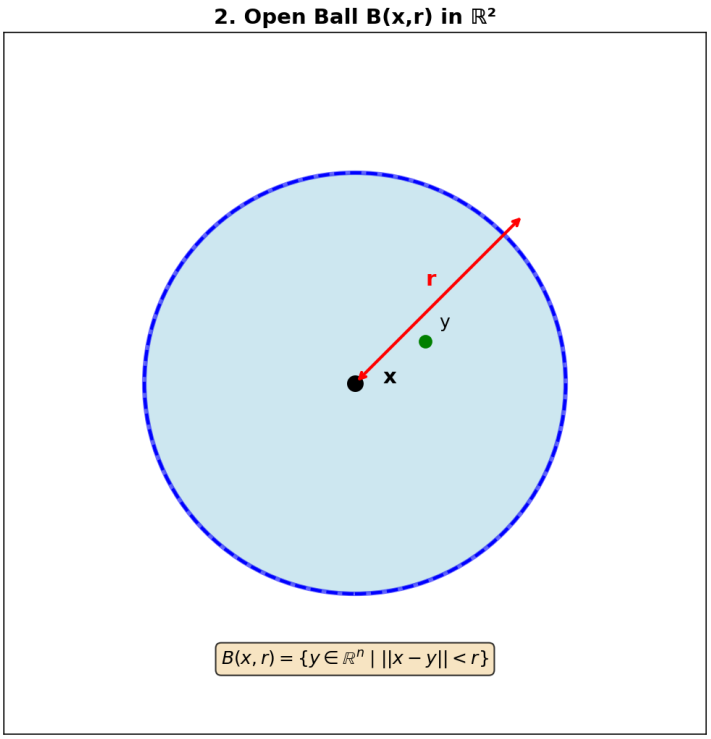


Figure 1.2: Open Ball $B(x, r)$ in $\mathbb{R}^2$

Logical Intuition: To guarantee that U is an open set, for any point x you find inside U, you must be able to inflate a sufficiently small "bubble" around x that remains entirely trapped within U.

1.4 Product Topology $(X \times Y)$

Suppose we have two separate topological spaces (X, T_X) and (Y, T_Y). If we combine them into a Cartesian product $X \times Y = \{(x, y) \mid x \in X, y \in Y\}$, we must define a new topology T_P for this joint space.

Instead of defining T_P from scratch, we elegantly induce it using a basis formed by the Cartesian product of the open sets of the parent spaces:

$$B = \{U \times V \mid U \in T_X, V \in T_Y\}$$

textbfSelf-Consistency Check in $\mathbb{R}^2$: If we multiply $\mathbb{R} \times \mathbb{R}$, the product topology basis creates open *rectangles*. However, the usual topology on $\mathbb{R}^2$ uses open *circular disks*. Are these two topologies different? No. They are perfectly equivalent. For any disk, you can embed a smaller rectangle inside it. Conversely, for any rectangle, you can embed a smaller disk inside it. Because these bases mutually enclose one another, they expand into the exact same topological space.

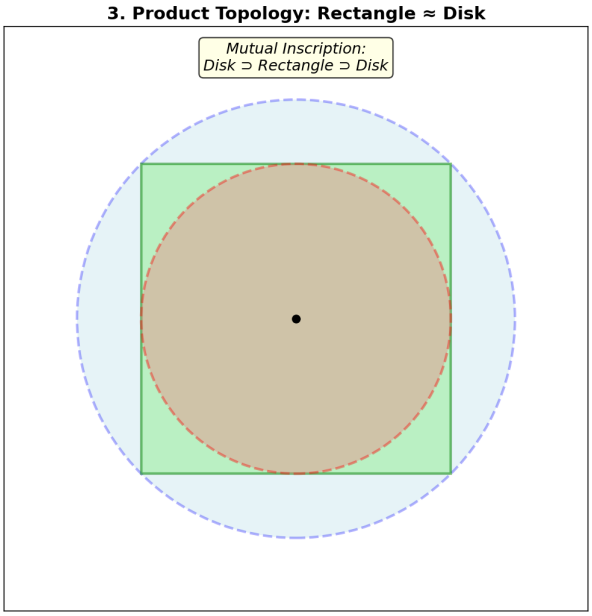
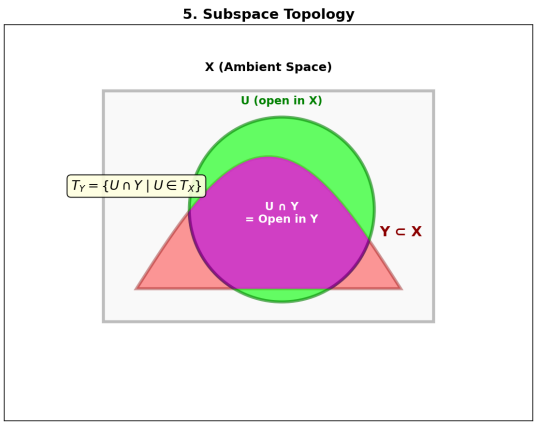


Figure 1.3: Product Topology: Rectangle $\subset$ Disk $\subset$ Rectangle $\Rightarrow$ Same Topology

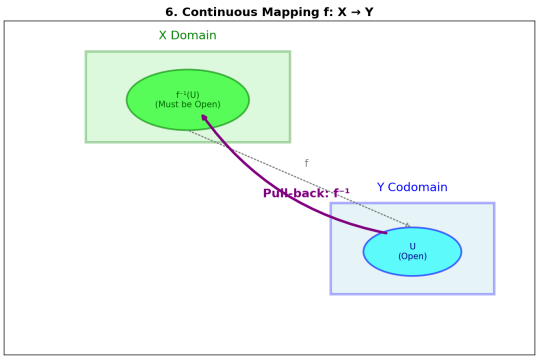
1.5 Subspace (Induced) Topology

In physics, we often only want to analyze a subset (like a specific 2D surface manifold) living inside a larger ambient space. Let $Y \subset X$, where X already has a topology T_X. We mathematically "borrow" the topology from X to define the topology T_Y on Y:

$$T_Y = \{U \cap Y \mid U \in T_X\}$$



(a) Subspace Topology



(b) Continuous Mapping

Logical Connection: Why do we use the intersection $U \cap Y$ instead of just saying "open sets in Y are the open sets in X"? Because if Y itself is not an open set in X, the fundamental axiom requiring the space itself to be open ($Y \in T_Y$) would fail. By taking the intersection with X itself (which is open in T_X), we evaluate $X \cap Y = Y$, safely guaranteeing that Y is an open set within its own subspace topology.

1.6 Continuous Mappings

In standard calculus, continuity is often loosely visualized as drawing a curve without lifting your pen. In topology, we define continuity rigorously and strictly through the behavior of open sets under inverse mappings (pull-backs).

Suppose we have two topological spaces (X, T_X) and (Y, T_Y). A mapping $f : X \to Y$ is **continuous** (denoted $f \in C^0$) if, for every open set U in the target space Y, its preimage in X is also an open set.

$$f^{-1}(U) = \{x \in X \mid f(x) \in U\}$$

textbfLogical Intuition: If you select an open interval in Y and map it backward to X, the resulting region must also be "open" (without rigid boundaries). If a function has a jump discontinuity, taking an open interval around the jump in Y and mapping it back to X will result in a set that includes a hard boundary (a closed end), directly violating the condition that the preimage must be open. Therefore, the function is logically not continuous.

1.7 Topological Equivalence: Homeomorphism

Once we have defined continuity, we can determine if two topological spaces are fundamentally the same. In topology, the absolute highest standard of equivalence between two spaces X and Y is called a **homeomorphism**.

Two spaces are homeomorphic if there exists a mapping $f : X \to Y$ that satisfies three strict conditions:

1. **Bijection:** The mapping is both injective (one-to-one) and surjective (onto).
2. **Continuous Mapping:** The function f is continuous.
3. **Continuous Inverse:** The inverse function f^{-1} exists and is also continuous.

Consequently, if these conditions are met, X and Y can be continuously deformed into one another without tearing or gluing.

1.8 Closed Sets

If an open set represents a region without its boundary, a closed set represents a region that includes its boundary. However, in topology, closed sets are defined entirely relative to open sets.

A subset $S \subseteq X$ is a **closed set** if and only if its complement, $X \setminus S$, is an open set.

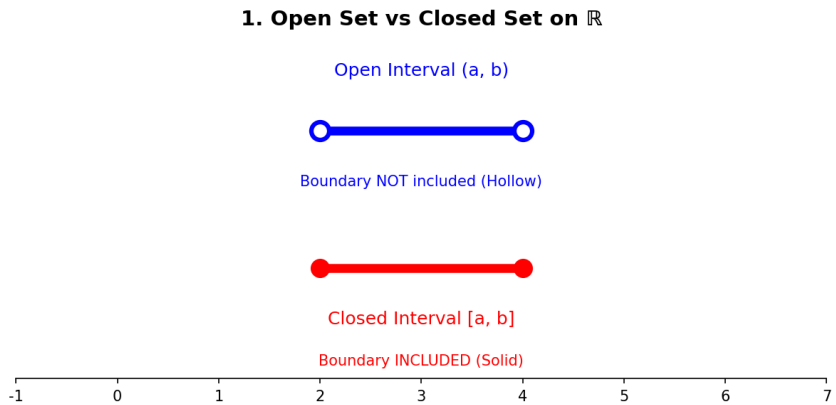


Figure 1.4: Open Set (a, b) vs Closed Set $[a, b]$

textbfExamples in the usual topology of $\mathbb{R}$:

- The open interval (a, b) is an **open set**.
- The closed interval $[a, b]$ is a **closed set** because its complement, $(-\infty, a) \cup (b, +\infty)$, is an open set.
- The half-open interval $[a, b)$ is **neither** open nor closed.
- The empty set $\emptyset$ and the entire space X are inherently **both** open and closed simultaneously.